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PE4218

SEAT No. : 15620
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[6554]-3010

S.Y. M.C.A. (Management)

MSD 615 MJ : MERN STACK DEVELOPMENT
(2024 Pattern) (Semester - III) (Elective - V)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) All questions are compulsory.
- 2) Figures to the right indicate full marks.
- 3) Draw diagrams wherever necessary.

Q1) Solve any two

[10]

- a) Implement a program using node js to handle asynchronous file operations using callbacks and promises.
- b) Create a basic HTTP server in Node.js that serves static files and handles GET requests.
- c) Write a code example that creates a custom event emitter, triggers an event named 'data Received' and has a listener function that handles it.

Q2) Solve any two

[10]

- a) Perform CRUD operations on a "Students" collection. Write MongoDB queries to :
 - i) Insert a new student document
 - ii) Update the grade of a student named "Raju" and
 - iii) Delete all students who have graduated
- X b) Design a complete MongoDB schema for a library management system with collections :
 - Books (title, author, ISBN, copies, genre)
 - Members (name, email, phone, membershipDate)
 - Transactions (book Id, member Id, issuedate, returndate, status)

P.T.O.

Write queries to :

- i) Add new book, member and transaction
- ii) Find all books borrowed by a member
- iii) Find overdue books
- c) Compare and contrast SQL and NOSQL databases. Justify why MongoDB is a good fit for a MERN stack application with a specific example.

Q3) Solve any two [10]

- a) Write express.js code to create routes for /home, /about and /contact.
- b) Write an express app to handle GET and post requests.
- c) Illustrate how to connect express with MongoDB to serve REST APIs.

Q4) Solve any one. [10]

- a) What are React Hooks? Describe the purpose of the useState and useEffect hooks with examples.
- b) Why is Redux used in React? Differentiate between local state and Redux.

Q5) Solve any one. [10]

- a) Create a "To-Do List" application using the MERN stack express and MongoDB should be used a backend API and React for frontend development.
- b) Develop a Restful API using express.js and MongoDB for managing product catalog. Display the product list using react as a frontend.

